



Challenges in Electricity Supply

Charging, Grid Capacity & How Voltaris Responds

Voltaris Mobility N.V. • Amsterdam, Netherlands • Founded 2016

“Drive the Current.”



The Electricity Challenge

The same dependence on electricity that gives Voltaris its opportunity also creates its hardest operational challenges. An electric vehicle is only as useful as the energy available to charge it — and delivering that energy, reliably and everywhere, is a significant undertaking. This document sets out the main difficulties and how the company addresses them.

Key Difficulties

- **Charging availability** — drivers need convenient, fast charging, especially those without home charging. Gaps in coverage create 'charging anxiety'.
- **Grid capacity** — large numbers of vehicles charging at once can strain local electricity grids, particularly at peak times.
- **Charging speed** — even fast charging takes longer than refuelling a petrol car, affecting long journeys.
- **Energy cost & source** — electricity prices vary, and the climate benefit depends on how clean the local grid is.
- **Rural & long-distance coverage** — building charging in low-traffic areas is costly but necessary for confidence.

How Voltaris Responds: VoltGrid

The company's primary answer is VoltGrid, its proprietary fast-charging network of 47,000 stalls. By building and operating its own network, Voltaris guarantees a baseline of reliable, fast charging for its customers and controls the experience end-to-end. Stalls are concentrated along major highway corridors and in city centres, and the network is increasingly opened to other brands' vehicles, improving utilisation and adding revenue.

Smart & Off-Peak Charging

To ease pressure on the grid, Voltaris vehicles and chargers support smart charging: shifting charging to off-peak hours when electricity is cheaper, cleaner and more abundant. This both lowers costs for customers and reduces the strain that simultaneous charging would otherwise place on local networks.

Vehicle-to-Grid and Storage

Rather than treating fleet batteries purely as a load on the grid, Voltaris is developing vehicle-to-grid capability, where parked vehicles can return energy to the grid at peak demand. Combined with stationary storage built from retired batteries, this turns the challenge of grid strain partly into a solution — the fleet helps balance the very grid it draws from.

Faster Charging & Better Range

On the vehicle side, the VoltCell solid-state battery platform is engineered for both long range — up to 780 km in the Orion — and fast charging, reducing how often and how long drivers must stop.

Each generation of VoltCell targets shorter charging times, directly attacking the speed problem.

Partnerships with Energy Providers

Finally, Voltaris works with utilities and renewable-energy providers to expand clean generation and grid capacity in step with EV growth, and to source low-carbon electricity for the VoltGrid network. The company is candid that the electricity challenge is real and ongoing, but argues that owning the charging network, investing in battery technology, and partnering with the energy sector together form a credible response.